

Particulate Matter Monitoring & Health Effects

Issue 2026-08-02 | Entry window 2 August 2026, plus an out-of-window sensing sweep | 31 records in scope

31

RECORDS IN SCOPE
(7 SUBTOPICS)

12

SENSING /
INSTRUMENTATION

6

ON SENSOR WATER /
HUMIDITY ARTEFACT

4

REJECTED WITH
LOGGED REASON

22

RECORDS WITH NO
HEALTH ENDPOINT

Signal of the day

Water is the day's unifying artefact — and one paper puts it inside the *reference* method, not just the low-cost sensor.

Six records this issue are about particle-bound water, and they stack into an argument. Chyzykhov et al. (*J Hazard Mater*) measured retained water directly by Karl Fischer coulometry with stepwise heating across 35 materials and an artificial-PM mixture: deliquescent inorganic salts retain most of it, water-to-solute ratio *falls* as filter loading rises, and PM-bound water releases across a thermal continuum rather than at a threshold. **Gravimetric PM mass — the quantity every low-cost sensor is calibrated against — is therefore composition-dependently biased by an operational definition of “dry”.** Against that, Guo et al. (*ACS ES&T Air*) attack the sensor side mechanistically, separating aerosol liquid water with EPA CSN speciation and correcting optics by Mie theory rather than fitting an RH polynomial; Kumpika et al. do fit the polynomial (PMS7003, $R^2 = 0.974$ against BAM) and find that **restricting duty cycle to 20 min/day materially slows drift**; Oybek et al. bound the raw penalty in a chamber — PM_{2.5} error rising from 1.2 to ~9 µg/m³ between 30% and 85% RH. Ginsburg et al. then show the consequence at scale: across 1,013 co-located PurpleAir units on six continents, US-derived corrections *do not transfer*, and the authors attribute this specifically to RH sensitivity and aerosol-composition variability. The practical reading for a distributed network is uncomfortable: an RH correction fitted against a gravimetric reference is partly fitting the reference's own water artefact, and that artefact is composition-specific — which is exactly what changes when you move the network.

What changed today

- ▶ **Calibration transferability now has numbers attached to distance, not just to geography.** Zhivkov et al. derive pollutant-specific uncertainty growth rates of 3.84–5.62% per km from a reference station by leave-one-station-out spatial CV over 24 months in Sofia, and convert them into a siting rule: co-location sites 6 km apart keep the midpoint penalty under 30%. This is the first record in this watch that gives a defensible answer to “how far apart can the anchors be” rather than asserting that co-location is necessary.
- ▶ **The high-exposure metro cohort came back close to null, and it matters.** Thies et al. estimate most London Underground commuters encounter hourly PM_{2.5} above 100 µg/m³ — 8–10× roadside — yet find only a 1.93% (0.58–3.66) survival deficit versus above-ground rail users, negligible cancer difference, **no dose-response**, and no meaningful benefit from a hypothetical 50% reduction. Either iron-rich mechanical-abrasion PM is substantially less toxic per unit mass than the ambient mixture, or the modelled cumulative exposure metric is too noisy to carry a gradient. Both readings are damaging to mass-only exposure surrogates.
- ▶ **Two independent groups attack the same station-density bias in satellite-derived PM_{2.5}.** Ding et al. show that models trained on urban-skewed monitor networks systematically overestimate rural concentrations — correcting it drops clean-site RMSE from 18.18 to 8.31 µg/m³ and national mean PM_{2.5} by ~55%, with the attributable-mortality burden in low-density areas revised down accordingly. Hu et al. approach the gap differently, filling AOD with WRF-Chem and injecting physics-derived features to reach hourly 1 km ($R^2 = 0.87$, RMSE 13.42 µg/m³). Both are arguments that the training-set geography, not the algorithm, is the binding constraint.
- ▶ **Near-source attribution improved on both the statistical and the deterministic side.** Burke et al. re-grid nonparametric trajectory analysis onto polar coordinates, raising estimated maximum near-road contributions by 22% for BC and 39% for NO_x — the previous Cartesian grid was smoothing away the near-field. Yu et al. resolve the street-canyon increment citywide for Hong Kong and find hotspot segments are only 12.2–18.8% of roads per district but expose ~184,000 residents in winter, with ΔC rising 19–26% from summer to winter.
- ▶ **EPA shipped tooling, not a paper.** Barkjohn et al. describe an R-Shiny application for sensor-versus-reference comparison and baseline QA/QC — explicitly not a correction method, but the kind of artefact that changes what a small group can operate.
- ▶ **Retraction to note.** *Sci Total Environ* retracted “Abundance and origin of fine particulate chloride in continental China” (2018;624:1041–1051), 10.1016/j.scitotenv.2026.182121. Check bibliographies.

Corpus analytics

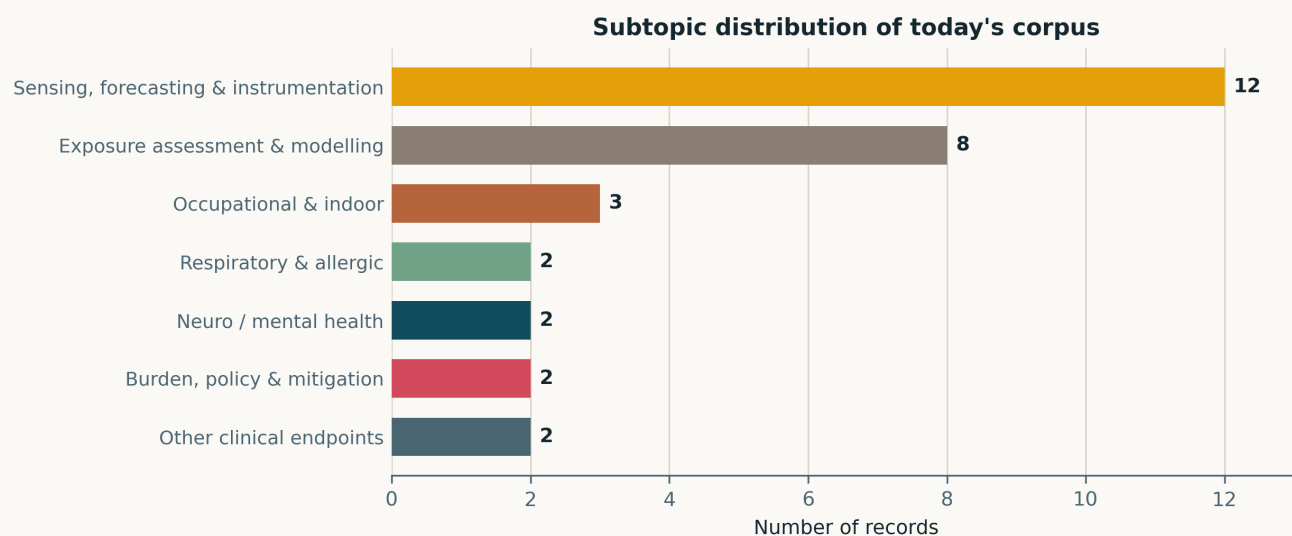


Fig. Subtopic assignment of the 31 in-scope records. The distribution is the most instrumentation-heavy of any issue to date (12/31, 39%), which is an artefact of method as much as of the field: the 2 August PubMed entry window was a single Sunday, so the deliberate Consensus sweep on calibration, humidity correction, siting and edge inference dominates. Health-endpoint clusters are correspondingly thin and no record this window sits in mechanistic toxicology, cardiovascular/metabolic or reproductive/developmental.

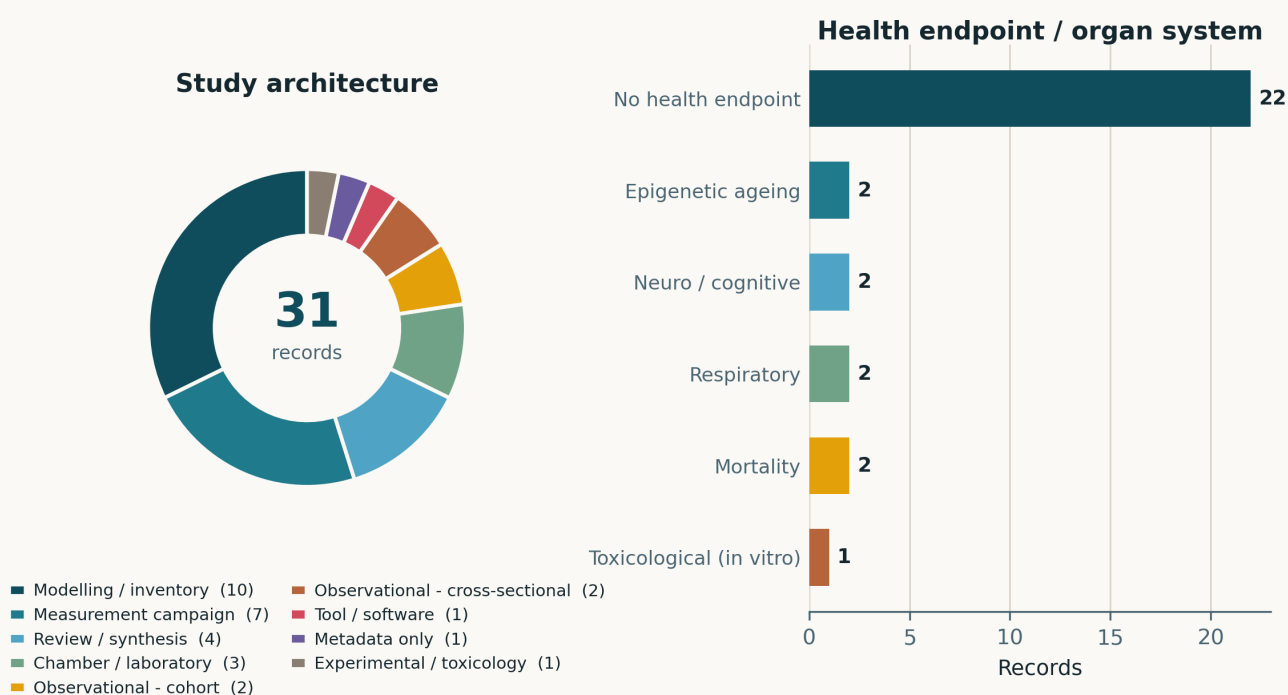


Fig. Left — study architecture. Right — health endpoint by organ system. **Twenty-two of 31 records carry no human health endpoint at all**, the highest share this watch has recorded. That is the direct signature of an instrumentation-weighted harvest, and it is the same siloing the 27 July issue flagged: the sensing literature and the epidemiology literature are still not citing each other's designs.

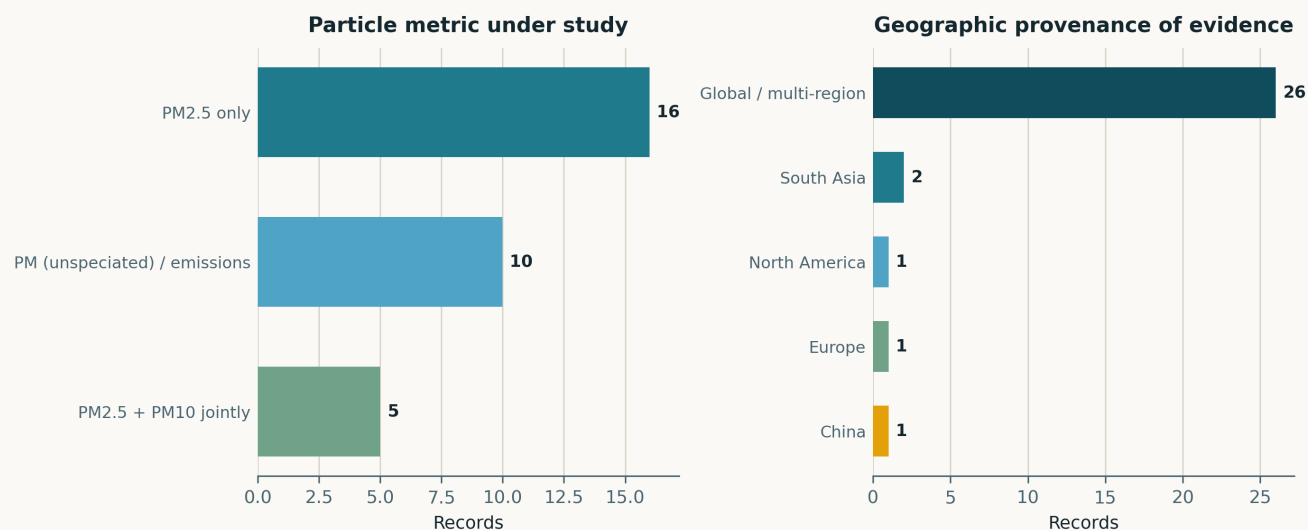


Fig. Left — particle metric. PM_{2.5} dominates; PM₁₀ appears jointly in the multi-pollutant calibration work only. Ultrafine particles are again absent as an explicit target, and no record measures particle number or size distribution as the primary quantity despite four papers whose stated mechanism (hygroscopic growth, Mie scattering) is size-resolved. Right — geographic provenance. Coverage is unusually wide this window (Europe, North America, East and South Asia), a side effect of the instrumentation sweep, which is less China-weighted than the health literature.

Life-course windows addressed today (records may span several stages)

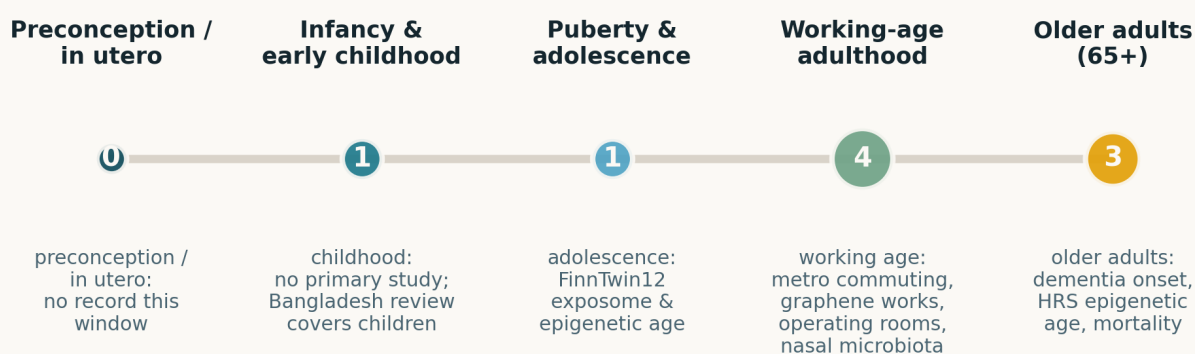


Fig. Life-course windows addressed. Counts are non-exclusive and small this window: working-age adults dominate through occupational and commuting settings, and there is **no in-utero record at all** — the first such gap since this watch began.

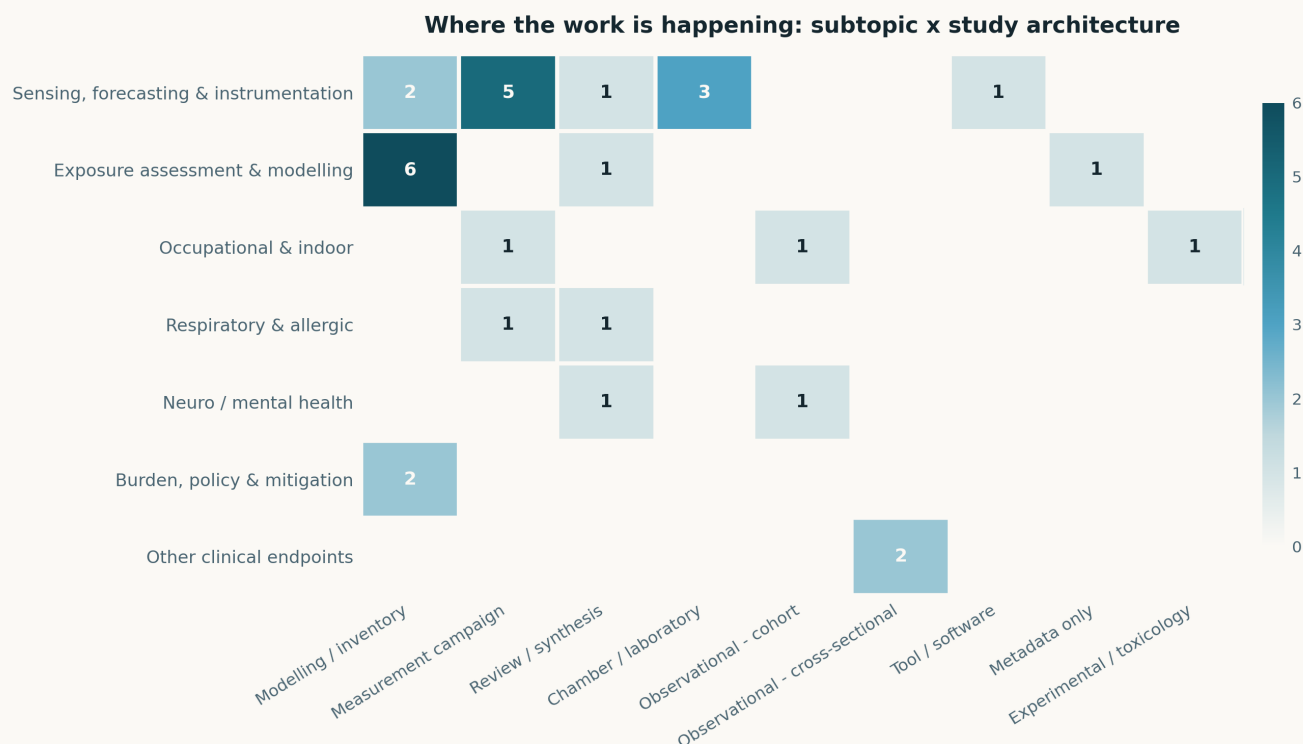


Fig. Subtopic × study-architecture cross-tabulation. The sensing row is concentrated in co-location, chamber and model-comparison designs with no health linkage, and the health rows contain no instrumentation component — with one exception worth naming: Kaur et al. pair workplace personal sampling with in vitro toxicity on the *same* collected material, which is the design template the rest of the corpus lacks.

No forest plot this issue — and that is the finding, not an omission.

Not one record in this window reports a ratio-metric effect estimate with a confidence interval. The closest is Thies et al., who give a survival difference of 1.93 percentage points (95% CI 0.58–3.66) between London Underground and above-ground rail commuters; plotted on the log-ratio axis used by every previous issue it would sit at 1.93 with the null line at 1.0 falling *inside* its interval, reading as a non-significant risk ratio when it is neither non-significant nor a risk ratio. It is therefore excluded rather than rendered misleadingly. Jahanshahi et al. report “clear” adjusted associations for PM_{2.5} and NO₂ with dementia onset but publish no hazard ratio in the abstract, and the remaining health records are reviews, microbiome compositional analyses or null SEM paths. This is the structural signature of an instrumentation-weighted harvest: the sensing literature reports R^2 , RMSE and nRMSE, and those do not belong on a forest plot. The pooled plot returns in the weekly rollup, which draws on a full week rather than a single day.

Paper-level digest

Ordered by cluster. Each entry gives the methodological core and the finding that matters, not an abstract paraphrase.

Sensing, forecasting & instrumentation

12 records

Chyzykhov et al. 2026 J Hazard Mater

Particle-bound water and the composition dependence of gravimetric PM mass

Karl Fischer coulometric titration with stepwise heating on 35 single-component chemicals and CRMs at two filter loadings (0.6 and 2.0 mg, summer- and winter-like), plus a compositionally realistic artificial PM mixture built both by sequential addition and in bulk. Deliquescent inorganic salts retain the most water; metals, metal oxides, PAHs and graphitic carbons sit near background. Water-to-solute molar ratio falls as absolute loading rises — compacted powder is less vapour-accessible — and temperature-resolved release clusters into three material groups differing mainly in strongly bound (250 °C) water. Bulk artificial-PM release profiles matched ambient PM₁ filters. **Limitation: powders are not ambient particles**; mixing state, morphology and coating history are not reproduced, so the transfer to real filters is argued from profile similarity, not demonstrated. For distributed sensing this is the uncomfortable upstream result — the mass reference itself carries a composition-dependent water term, so an empirical RH correction fitted against it is partly correcting the reference.

doi: 10.1016/j.jhazmat.2026.143142

Ginsburg et al. 2026 Aerosol Sci Technol

Do US PurpleAir corrections travel? 1,013 co-located sensors on six continents

Global evaluation dataset, January 2023–March 2025, of PurpleAir units co-located within 500 m of reference-grade monitors, assessed at 24-h and 1-h averaging on R^2 , RMSE, nRMSE, precision (CV) and AQI category correctness (24-h and NowCast). Raw PurpleAir overestimates PM_{2.5} in most regions and fails EPA sensor performance targets outright. The three widely used US correction equations improve accuracy but **transfer poorly outside North America**; a global quadratic helps in Australia and New Zealand and hurts elsewhere; **regional calibrations consistently win**, meeting EPA 24-h thresholds in several regions and lifting 24-h AQI correctness to $\geq 77\%$. **Limitation: 500 m is a generous co-location radius**, so some of the residual regional error is spatial mismatch rather than sensor bias, and the regional strata are unevenly populated. Directly actionable — it says region-specific correction is mandatory wherever a co-location exists, and quantifies what is lost when it does not.

doi: 10.1080/02786826.2026.2676293

Zhivkov et al. 2026 Atmosphere

How fast does a calibration decay with distance from its reference station?

Twenty-four months of hourly data (Aug 2023–Jul 2025) from Sofia: five official reference stations and 22 AirThings low-cost units, four co-located. Random Forest calibration achieved R^2 0.53–0.75 across PM_{2.5}, PM₁₀, NO₂ and O₃, improving on multiple linear regression by 40% (O₃) to 408% (PM_{2.5}). Leave-one-station-out spatial cross-validation yields pollutant-specific uncertainty growth rates α of **3.84–5.62% per km** (significant for PM₁₀ and O₃ only, $p < 0.05$), applied to 18 non-co-located sensors to generate 1.2 M calibrated hourly values with 95% prediction intervals. **Limitation: α is estimated from five stations in one city**, so its significance for PM_{2.5} is not established and the linear-in-distance form is an assumption. Still the most directly usable network design result in the window: 6 km anchor spacing keeps the midpoint uncertainty increase below 30%.

doi: 10.3390/atmos17040335

Guo et al. 2026 ACS ES&T Air

Mechanistic RH correction by separating aerosol liquid water from dry mass

Two-stage framework for a nephelometer plus optical particle counter: (i) Mie-theory optical calibration accounting for size distribution, refractive index and hygroscopic growth; (ii) quantification of aerosol liquid water mass from EPA Chemical Speciation Network composition to recover dry-equivalent PM_{2.5} at high RH. Corrected values agree closely with EPA reference measurements, and the recovered ALW is itself useful for aqueous-phase aerosol chemistry. Practical guidance is given for sites without co-located CSN speciation. **Limitation: the method is only as good as the speciation surrogate** — outside the CSN footprint the ALW term becomes an assumption, and the paper does not report how much accuracy is lost when the surrogate is wrong. This is the right-shaped answer to the humidity problem for anyone unwilling to refit an empirical curve at every new site.

doi: 10.1021/acsestair.5c00225

Yaqoob et al. 2026 IEEE Access

Calibration that survives a change of city and a change of reference instrument

PurpleAir and Alphasense OPC-N3 in Antwerp, Oslo and Zagreb, hourly PM_{2.5} and PM₁₀, calibrated with Random Forest, a compact feedforward network, and HybriCal-AQ (a hybrid NN taking a standardised out-of-fold RF hint). Models are evaluated separately against TEOM, FIDAS and generic reference grades to mirror real reference availability, with a train-first split/impute/encode pipeline to avoid leakage. HybriCal-AQ wins on PM_{2.5} (Oslo-FIDAS/PurpleAir RMSE 1.79 → 1.18, R^2 0.94 → 0.97) while **plain RF usually remains better for PM₁₀** (Antwerp 4.76 vs 6.58). A sparse-reference test withholding 25% of reference observations probes robustness. **Limitation: three cities is not a transferability test** — every model is still trained and evaluated within-city, so this bounds instrument sensitivity, not spatial generalisation. The explicit reference-instrument stratification is the transferable methodological idea.

doi: 10.1109/access.2026.3656120

Kumpika et al. 2026 Environ Pollut Bioavail

Humidity and ageing corrected together, and duty cycle as a drift control

PMS7003 units in Chiang Mai across varying RH, validated against a government BAM and verified in dry air with a diffusion dryer. Polynomial corrections up to third order gave strong agreement at elevated humidity ($R^2 = 0.9737$). The distinctive result is operational rather than statistical: **limiting operation to 20 minutes/day measurably reduced long-term performance drift** relative to continuous running. **Limitations: single site, single sensor model, single aerosol regime** (Chiang Mai's biomass-burning season is compositionally atypical), and the duty-cycle finding is confounded with total particle throughput — a sensor that samples less sees less mass through its optics, so “ageing” and “fouling” are not separated. Worth replicating; if it holds, duty cycling is a nearly free lifetime extension for battery-constrained nodes.

doi: 10.1080/26395940.2026.2616098

Qian et al. 2026 Atmos Meas Tech

AutoML calibration staged through an intermediate drift-reference sensor

Multi-stage indoor calibration framework linking deployed low-cost sensors to intermediate drift-correction reference sensors and finally to a reference-grade instrument, with separate models fitted for clean and pollution-event concentration ranges. Evaluated in a controlled chamber on

two sensor models under diverse indoor sources and uncontrolled ambient conditions: $R^2 > 0.90$, NRMSE and sMAPE roughly halved against uncalibrated data, bias effectively removed. **Limitation: chamber conditions and an indoor concentration range**; the two-regime split is fitted, not physically motivated, and nothing establishes that the intermediate-reference tier is worth its cost against direct periodic co-location. The tiered-reference architecture is nonetheless the interesting part — it is a plausible answer to maintaining a large network with few reference-grade instruments.

doi: 10.5194/amt-19-603-2026

Oybek et al. 2026 Am J Appl Sci Technol

Chamber bounds on the temperature and humidity penalty, by sensor chemistry

Controlled-chamber factorial over six temperature (15–35 °C) and relative humidity (30–90%) combinations for an optical PM_{2.5} counter, a metal-oxide NO₂ sensor and an electrochemical CO₂ sensor. PM_{2.5} error rises from 1.2 µg/m³ at 30% RH to ~9 µg/m³ at 85% RH; a simple linear correction recovers up to 74% of it. **Limitations are substantial**: one unit per chemistry, no replication, an unstated test aerosol, a venue with no visible peer-review record, and a linear correction applied to a process (hygroscopic growth) that is not linear. Carried as a magnitude anchor only — the number is consistent with better studies and useful for sizing an error budget, but nothing here should be cited as a correction method.

doi: 10.37547/ajast/volume06issue05-25

Yan 2026 Sensor Review

What 90+ ML calibration studies actually agree on

Structured review of more than 90 peer-reviewed studies (2013–early 2024) across optical, electrochemical, metal-oxide and NDIR sensors and across PM, O₃ and NO₂. Consensus findings: ML calibration cuts error by >50% and lifts R^2 above 0.8–0.9; **optical PM sensors post-calibration cluster at R^2 0.8–0.95 with slopes near unity under long-term co-location**, while electrochemical NO₂/O₃ sit at 0.6–0.9 with much larger site-to-site spread. Neural networks and gradient boosting beat simple regression when data are plentiful; regression is more stable when they are not. Persistent problems named: drift, absent standard protocols, poor cross-site generalisation, and regulator resistance to black-box models. **Limitation: the search ends in early 2024** and the review is narrative in its weighting, with no quantitative pooling. Useful as a citation anchor for the claim that transferability, not fit, is the open problem.

doi: 10.1108/sr-10-2025-0761

Burke et al. 2026 Atmos Environ

Polar-grid nonparametric trajectory analysis recovers near-field source structure

Nonparametric trajectory analysis combines continuous concentrations with local-scale back trajectories from wind speed and direction to map source-area impact. The regular Cartesian grid plus kernel smoothing used previously suppresses spatial variability exactly where a near-source monitor needs it; re-gridding onto polar coordinates gives finer resolution close to the site. Tested on one year of continuous data from a near-road network site in St. Louis. Areas nearest the roadway contributed most (1.4 µg/m³ BC, 39 ppb NO_x), and **polar NTA raised maximum estimated contributions by 22% (BC) and 39% (NO_x)**. Background-subtracted PM_{2.5} tracked the traffic-related species but also revealed nearby point-source contributions. **Limitation: one site, one year, no independent validation** of the recovered source field — the improvement is demonstrated as internal consistency, not against known emissions. Immediately relevant to siting and to interpreting single-node data in a sparse network.

doi: 10.1016/j.atmosenv.2026.121951

Barkjohn et al. 2026 Atmosphere

An R-Shiny application for sensor-versus-reference QA/QC

Tool paper describing an application that lets air-quality agencies (i) assess sensor data quality against nearby regulatory reference monitors, (ii) apply baseline QA/QC and (iii) characterise local conditions, with reproducibility as an explicit design goal. Demonstrated on two cases: PM_{2.5} sensor performance under Midwestern wildfire smoke, and a year of O₃ sensor performance. The authors are explicit that advanced correction remains out of scope. **Limitation: no new method and no quantitative evaluation of the tool itself** — the case studies illustrate rather than validate. Carried because availability, not novelty, is what determines whether a small group's sensor data ever gets screened at all; this lowers that barrier and standardises the first pass.

doi: 10.3390/atmos16111270

Reis 2026 Sensors

A low-power node that does its own inference and stops talking so much

Design and system-level optimisation of an embedded node integrating PM, gas, temperature and humidity sensing on a low-power microcontroller with a neural inference accelerator, using adaptive duty-cycling, hierarchical power domains and edge-level data reduction. On-device quantised fixed-point models perform three-class normal/anomalous/critical classification. Reported: **94% inference accuracy, 0.87 ms latency, ~2.9 mWh average power**, and an adaptive LoRaWAN strategy cutting transmissions by ~88% versus periodic reporting, enabling solar-battery autonomy. **Critical limitation: every number comes from MATLAB/Simulink simulation**, not a deployed node — so the accuracy figure describes a synthetic label distribution and the power figure omits the real cost of sensor warm-up, radio retries and thermal effects. Read as an architecture proposal, not a measurement.

doi: 10.3390/s26020703

Exposure assessment & modelling

8 records

Ding et al. 2026 Geo-spatial Inf Sci

Station density is the bias: urban–rural balanced PM_{2.5} estimation

Satellite AOD-based PM_{2.5} retrieval is trained on ground networks that are heavily urban, and the resulting models overestimate rural concentrations. This framework integrates AOD, reanalysis fields and ground measurements inside an ML model guided by data-assimilation principles, adaptively constraining non-urban estimates with low-weighted reanalysis samples while preserving urban fidelity, then ensemble-averaging over dynamically weighted background–observation settings. Applied to China 2015–2021 it cut clean-site RMSE from **18.18 to 8.31 µg/m³**, lowered the national mean PM_{2.5} by ~55%, and substantially corrected overestimated PM_{2.5}-attributable mortality in low-density areas, revealing a sharper urban–rural decline gradient than previously reported. **Limitation: the rural “truth” is partly reanalysis**, so the correction is validated against a small number of genuinely clean sites and the 55% national revision rests on that thin anchor. If it survives replication it is a large downward correction to a widely cited burden estimate.

doi: 10.1080/10095020.2026.2624335

Hu et al. 2026 Geo-spatial Inf Sci

Hourly 1 km seamless PM_{2.5} by filling AOD gaps with a chemical transport model

Rather than stacking ground, satellite and CTM outputs, this framework first builds seamless AOD by merging WRF-Chem AOD with satellite retrievals, then injects physics-informed features — the temporal component of $\text{PM}_{2.5}$ and the vertical structure of AOD from WRF-Chem — into a geospatially aware LightGBM that also exploits monitoring-site spatial structure. Site-based cross-validation gives R^2 0.87, RMSE 13.42 $\mu\text{g}/\text{m}^3$, MAE 8.68 $\mu\text{g}/\text{m}^3$, with day-and-night 1 km hourly coverage demonstrated through a pollution episode. **Limitation: site-based CV overstates performance for locations far from any monitor**, which is precisely where seamless products are used; no leave-region-out test is reported. The gap-filling-first ordering is the transferable design choice — it removes the missing-AOD selection effect that biases most fusion products toward clear-sky, low-pollution conditions.

doi: 10.1080/10095020.2026.2680855

Yu et al. 2026 Environ Pollut

Street-canyon increment to near-road NO_2 , resolved for a whole city

Coupled WRF-CMAQ-ADMS framework with a purpose-built street-scale traffic emission inventory for Hong Kong, run in paired configurations with and without the street-canyon module to isolate the canyon-induced increment ΔC . Hotspots (segments with $\Delta\text{C} > 5 \mu\text{g}/\text{m}^3$, roughly a 1% increment in long-term NO_2 -attributable added risk) account for 12.2–18.8% of roads in the four worst districts. Canyon effects are seasonal — mean hotspot ΔC rises 19–26% from summer to winter under weaker ventilation — and about 184,000 residents are exposed to substantial winter increments. **Limitations: NO_2 only, no PM increment reported**, and the canyon module is validated indirectly through the coupled system rather than against canyon-resolved measurements. The reason to read it here is siting: it identifies, at segment resolution, where a sparse regulatory network is structurally blind and where dense low-cost deployment would actually add information.

doi: 10.1016/j.envpol.2026.128871

Nguyen et al. 2026 Eng Res Express

Systematic review of satellite-plus-ML $\text{PM}_{2.5}$ prediction, 2015–2025

Synthesis of a decade of work integrating satellite products — AOD from low-Earth-orbit and geostationary platforms, top-of-atmosphere reflectance, Ångström exponent — with machine and deep learning. Auxiliary meteorological, land-use, demographic and infrastructural covariates are identified as the main lever on predictive bias, and data normalisation plus missing-AOD handling as the main determinants of model stability. Random Forest remains a robust benchmark; hybrid and transformer architectures gain accuracy by capturing spatiotemporal dependence. The review's forward-looking section is its most useful part: spatiotemporal explainable AI and physics-guided learning as the route out of the transferability problem. **Limitation: narrative synthesis with no pooled quantitative comparison**, so “better accuracy” is asserted across studies with incomparable validation designs — the same site-based-CV optimism that inflates the individual papers.

doi: 10.1088/2631-8695/ae7028

Kim et al. 2026 Environ Pollut

Wildfire plumes can lower downwind sulfate by scavenging OH

CMAQ driven by the high-resolution FINN fire inventory, applied to the July 2014 Siberian wildfire episode. Long-range transport generally raised particulate matter across Northeast Asia, but during 26–29 July an anomalous sulfate decrease of up to $-1.56 \mu\text{g m}^{-3}$ appeared over Northeast China, the Shandong Peninsula and the Yellow Sea. The mechanism is competitive: fire-emitted CO and VOCs scavenge hydroxyl radicals, suppressing SO_2 oxidation. Source apportionment attributes 92.30% of the suppressed sulfate to inland anthropogenic SO_2 and 7.70% to oceanic sources. **Limitations: a single episode, model-only, with the mechanism inferred from the chemistry solver rather than measured OH**; FINN emission uncertainty propagates directly into the sign of the result. Relevant because it breaks the assumption that fire contributions to PM are additive — a source-apportionment scheme that assumes additivity will misattribute here.

doi: 10.1016/j.envpol.2026.128872

Parajuli et al. 2026 Atmos Environ

Which soil NO_x parameterisation you pick changes modelled ozone and PM

As fossil-fuel NO_x declines, ozone becomes more sensitive to non-fossil nitrogen, yet most air-quality models still use simple soil- NO parameterisations. This study compares Yienger & Levy and BDSNP against the mechanistic FEST-C* fertiliser emissions tool, propagating each through CMAQ over the United States and evaluating against rural NO_2 monitors and satellite NO_2 columns. FEST-C* agricultural soil NO/HONO was double YL and two-thirds of BDSNP, improved agreement at some rural sites, and had intermediate correlation but the lowest bias over agricultural areas. **Limitation: the PM response is reported only qualitatively** — the paper is NO_x - and ozone-centred, so a PM-focused reader gets the direction and not the magnitude. Carried because it identifies a structural uncertainty in a model layer that PM exposure surfaces routinely inherit without examination.

doi: 10.1016/j.atmosenv.2026.122225

Roky et al. 2026 Environ Pollut

Distance-resolved lead dispersion and crop uptake from open battery-plate burning

Controlled field burns of lead-acid battery plates in Kushtia, Bangladesh (three plates per event, $149.7 \pm 4.5 \text{ g}$ total mass loss), monitored on a triangular sampling grid to 45 m with simultaneous $\text{PM}_{2.5}/\text{PM}_{10}$, passive Pb deposition on filter papers, and Pb accumulation in red amaranth. All three response variables showed strong distance-dependent gradients with significant distance, week and interaction effects ($p < 0.0001$); $\text{PM}_{2.5}$ tracked both deposition and crop Pb more strongly than PM_{10} , and background plots stayed below detection, supporting atmospheric deposition as the dominant pathway. Crop-response model $R^2 = 0.948$. **Limitation: the threshold distance is not a single number** — the Codex leafy-vegetable benchmark is fresh-weight and the plant data dry-weight, so it had to be evaluated across a plausible moisture range, and 45 m does not bound the plume. A clean demonstration that a short-range $\text{PM}_{2.5}$ gradient is a usable proxy for a metal deposition field.

doi: 10.1016/j.envpol.2026.128853

Lu et al. 2026 Atmos Environ

Moving-vehicle wakes and noise barriers in traffic emission dispersion

Metadata-only record. Surfaced by the Crossref by-journal leg as a new *Atmospheric Environment* deposit dated 2 August 2026; Elsevier deposits no abstract to Crossref and the article is not yet in PubMed or Europe PMC, so **no summary is offered and none should be inferred from the title**. Listed so that the window's coverage is complete and so the record can be picked up and summarised when an abstract becomes available. Authors: Lu, Li, Wang, Wu, Liu, Yang, Zhao, Zheng. The topic — vehicle-induced turbulence interacting with barrier geometry — is squarely relevant to near-road sensor siting, which is why it is carried rather than dropped.

doi: 10.1016/j.atmosenv.2026.122256

Occupational & indoor

3 records

Thies et al. 2026 Int J Popul Data Sci

The London Underground: very high $\text{PM}_{2.5}$, very little dose–response

Two retrospective cohorts from the ONS Longitudinal Study of economically active adults living and working in London (1991, 2001, 2011 Censuses). Cohort 1 ($n = 7,343$) compared Underground commuters with above-ground rail users; Cohort 2 ($n = 4,094$) tested dose-response against externally linked cumulative LU $PM_{2.5}$ built from Transport for London data, $PM_{2.5}$ measurements and geospatial modelling. Outcomes were all-cause mortality (1991–2017) and cancer incidence (1991–2015), analysed with inverse probability weighting and the parametric g-formula. Most LU commuters were estimated to encounter hourly $PM_{2.5}$ above $100 \mu g/m^3$ ($8\text{--}10\times$ roadside), yet the survival deficit was **1.93% (95% CI 0.58–3.66)**, cancer differences negligible, **no dose-response** emerged, and a hypothetical 50% $PM_{2.5}$ reduction barely moved survival. **Limitation: exposure is modelled, not measured on individuals**, and census-derived commute mode is a coarse, decadal proxy — non-differential misclassification of this size can flatten a real gradient. Either way, iron-rich mechanical-abrasion PM does not behave like ambient mass.

doi: 10.23889/ijpds.v11i5.3705

Kaur et al. 2026 NanoImpact

Workplace graphene: the material you make is more toxic than the air you breathe

Pilot study pairing occupational exposure assessment in an operating graphene production facility with in vitro toxicology on the very material collected there. Stationary and personal samplers fed elemental carbon analysis, Raman microscopy-spectroscopy (which confirmed graphene in the collected PM), and viability testing in A549 and BEAS-2B cells. **Personal EC ranged $3\text{--}484 \mu g/m^3$** , peaking during graphite powder handling. The dissociation is the finding: **workplace-collected PM was not cytotoxic at tested concentrations, while dispersions of as-produced liquid-phase-exfoliated graphene caused dose-dependent viability loss**. **Limitations: one facility, pilot scale, two cell lines, acute viability only**, and collected-sample doses may simply be below any threshold rather than genuinely benign. Nonetheless the design — sample the workplace, then test that sample rather than a pristine surrogate — is the coupling this field keeps asking for.

doi: 10.1016/j.impact.2026.100642

Nath et al. 2026 J Anaesthesiol Clin Pharmacol

Operating-room particulate: HEPA helps, cautery undoes it

Portable monitoring of CO_2 , TVOC, formaldehyde, $PM_{2.5}$ and PM_{10} across operating rooms with and without HEPA filtration. HEPA rooms were significantly cleaner ($PM_{2.5}$ 19.78 ± 8.81 vs $29.83 \pm 4.56 \mu g/m^3$, $p = 0.026$; PM_{10} 29.93 ± 12.96 vs 38.61 ± 6.89 , $p = 0.04$) while other pollutants were comparable. Electrocautery raised TVOC, $PM_{2.5}$ and PM_{10} ; ORs stayed well below the control area in winter ($PM_{2.5}$ 17.0 vs $36.66 \mu g/m^3$); average CO_2 was often high enough to matter for staff. **Limitations: small, single-centre, cross-sectional, with no personal sampling and no health endpoint**, and the HEPA and cautery comparisons appear to reuse overlapping room-groups, so the two effects are not cleanly separated. Included as a reminder that the highest-value indoor PM measurements are often minute-resolved and event-driven, which is where low-cost nodes are strongest.

doi: 10.4103/joacp.joacp_380_25

Respiratory & allergic

2 records

Solazzo et al. 2026 Hyg Environ Health Adv

Season and pollutant interact on the nasal microbiome

Longitudinal repeated-measures design: 26 healthy Italian adults sampled weekly for three weeks in winter and three in summer, with 16S rRNA sequencing (124 samples) and whole-genome shotgun sequencing (141 samples), against weekly indoor TSP and outdoor particulate matter, black carbon, benzene and CO. The microbiota was stable within season but differed between seasons, with winter enrichment of *Moraxella* — notably among women with children. Higher pollutant levels were negatively associated with commensal abundance, particularly *Corynebacterium*, and **significant season $\times$ pollutant interactions** appeared: in summer, *Staphylococcus epidermidis* and *Cutibacterium granulosum* were negatively associated with PM. Sex explained the most variance in diversity. **Limitations: $n = 26$ with many taxa tested**, exposure assigned from area monitors rather than personal sampling, and season is collinear with everything from heating to indoor occupancy. Hypothesis-generating; the interaction structure is the part worth chasing with a larger cohort and personal monitors.

doi: 10.1016/j.heha.2026.100189

Haque et al. 2026 Health Sci Rep

PM and respiratory disease in Bangladesh: what 42 studies show

Narrative review of 42 studies (2013–2025) drawn from seven databases on fine and ultrafine PM and gaseous pollutants and respiratory outcomes in Bangladesh. $PM_{2.5}$ in Dhaka, Chattogram and Khulna is reported at **$8\text{--}18\times$ WHO guideline levels**, with persistent hotspots around brick kilns, industrial zones, transport corridors and seasonal biomass burning, and consistent associations with asthma, COPD, bronchitis, acute respiratory infection and hospital admission, concentrated in children, older adults and occupational groups. **Limitations: narrative, not systematic** — no risk-of-bias assessment, no pooling, and heavy reliance on primary studies with sparse monitoring infrastructure, so the exposure side of most included work is itself weak. Carried because it documents where the burden is largest and the measurement network is thinnest, which is the standing argument for low-cost network deployment in South Asia.

doi: 10.1002/hsr2.72925

Neuro / mental health

2 records

Jahanshahi et al. 2026 Int J Popul Data Sci

Dementia onset and long-term $PM_{2.5}$ in a genuinely low-pollution setting

Northern Ireland census data linked to administrative prescription records, 2010–2016, with dementia onset proxied by first dispensed dementia medication — an endpoint that avoids diagnostic-coding variability at the cost of missing untreated cases. Cox proportional hazards models estimated the effect of five-year moving-average $PM_{2.5}$ and NO_2 with extensive confounder adjustment. **Clear positive associations in both unadjusted and adjusted models**, persisting across age and sex subsamples and more tentative below age 70. **Limitations: the abstract reports no numerical hazard ratios or intervals**, so the effect size cannot be entered into the forest plot; prescription-based ascertainment under-counts and lags true onset; and exposure is residence-assigned with no mobility term. The contribution is contextual — it extends the dose-response into a low-concentration range where a threshold has repeatedly been proposed and not found.

doi: 10.23889/ijpds.v11i5.3644

Lin et al. 2026 Genes Dis

A lung-brain axis hypothesis for pollution-associated epilepsy

Hypothesis-driven review proposing that inhaled air pollutants disrupt lung microbiome homeostasis and that this dysbiosis — by analogy with the gut-brain axis — contributes to epileptogenesis. Assembles three strands: environmental associations between air pollution and epilepsy

susceptibility, links between lung-intrinsic microbiota dysregulation and neurological dysfunction, and candidate molecular, immunological and neural pathways by which the lung–brain axis could modulate seizure threshold. **This is a proposal, not evidence.** No new data; the pollution–epilepsy associations it rests on are sparse and confounded, and no step of the chain (PM → lung dysbiosis → neural excitability) has been demonstrated in the same system. Carried at tier C because the mechanistic-toxicology cluster has been empty this window and because it names testable intermediate endpoints — lung microbiome composition after controlled exposure is measurable now.

doi: 10.1016/j.gendis.2026.102240

Burden, policy & mitigation

2 records

Kim et al. 2026 J Environ Manage

Full diesel-to-BEV replacement: where the benefit lands is not where the people are

CMAQ counterfactual in which all diesel vehicles are replaced by battery electric vehicles, evaluated with population-weighted concentration across urban core, urban periphery and rural strata — an explicit attempt to move policy appraisal off regional-average concentration change. Despite large on-road NO_x declines in the urban core and periphery, **annual daily-maximum 1-h ozone rose there through weakened NO titration** while falling in rural areas, so ozone PWC increases were concentrated where people live. **PM_{2.5} moved the opposite way:** the largest concentration reductions occurred rurally, where low population density limited the exposure benefit. The result is a genuine spatial-equity tension inside a policy usually scored as unambiguously good. **Limitations: an idealised full-replacement scenario, one domain, one meteorological year, no fleet-turnover or grid-emissions accounting, and no health-endpoint monetisation.**

doi: 10.1016/j.jenvman.2026.130597

Gardner-Frolick et al. 2026 iScience

Double disproportionality: few emitters, and they sit next to marginalised people

National analysis asking two questions jointly — which industrial subsectors have emissions concentrated in a few facilities, and whether those facilities are disproportionately sited near structurally marginalised groups. Emission disproportionality holds across pollutants and subsectors, and siting patterns diverge by settlement type: **oil and gas extraction disproportionately affects marginalised urban populations, mining and quarrying rural ones.** Urban Indigenous populations are strongly affected by both all-emitter and high-emitter siting, and low-income and Black populations are more likely to live near *clusters* of emitting facilities — a cumulative-hazard framing that single-facility analyses miss. **Limitations: proximity is not exposure** (no dispersion modelling, no monitoring), self-reported inventory emissions, and areal-unit demographics vulnerable to the modifiable areal unit problem. The cluster result is the one that should drive monitor siting.

doi: 10.1016/j.isci.2026.116827

Other clinical endpoints

2 records

Opperbebeck et al. 2026 Environ Int

Exposome-wide search for epigenetic ageing — and how much is still missing

FinnTwin12: 847 participants measured at ages 12, 14, 17 and 22, with 186 unique exposures spanning lifestyle, green environments, air pollutants and demographics, analysed by exposome-wide association plus knockoff boosted trees, sNPLS and Boruta to allow non-linearity and interaction. Epigenetic age at 22 (PCGrimAge, DunedinPACE) was **~28% explained** (R^2 25.7% and 30.8%). Smoking, alcohol, youth unemployment and green-space *size* predicted accelerated ageing; tree cover, vegetation index, neighbourhood age structure and **aerial black carbon predicted decelerated ageing** — a sign reversal that the authors do not resolve and that almost certainly reflects urban socioeconomic confounding rather than a protective effect of BC. Twin modelling attributes the missing variance mainly to environmental factors unshared between co-twins. **Limitations: one cohort, modest n for 186 exposures, and BC assigned at residence.** Read for the missing-exposome decomposition, not the BC coefficient.

doi: 10.1016/j.envint.2026.110430

DeAngelis et al. 2026 Soc Sci Med

Residential context and epigenetic ageing: a null for air pollution

Health and Retirement Study participants providing venous blood in 2016 ($n = 3,187$, mean age 69), with multigroup structural equation models decomposing paths between a latent epigenetic-age-acceleration variable (GrimAge, PhenoAge, DunedinPoAm38) and five 2010 residential features: racial clustering, air pollution, greenspace, household income and perceived disorder. White respondents in areas with high White clustering showed lower EAA, about 25% of it accounted for by higher area household income; **among Black participants essentially no residential path reached significance, and air pollution was not an independent predictor in either group.** **Limitations: single-timepoint methylation, exposure indexed to areal units six years before assay, and a null that is as consistent with measurement error in the area-level pollution term as with absence of effect.** Carried as a negative result: coarse area-level pollution proxies are not resolving what individual-level exposure assessment might.

doi: 10.1016/j.socscimed.2026.119635

Corpus register

First author	Focus	Journal	Design	Metric	Region
Neuro / mental health					
Jahanshahi et al.	Long-term PM _{2.5} and NO ₂ and dementia onset in a low-pollution setting	Int J Population Data Science	Registry cohort + geospatial	PM _{2.5} , NO ₂ (5-yr moving average)	Northern Ireland
Lin et al.	Lung-brain axis and lung dysbiosis as a route from inhaled pollutants to epileptogenesis	Genes & Diseases	Critical review	Inhaled air pollutants (un-specified)	n/a
Respiratory & allergic					

First author	Focus	Journal	Design	Metric	Region
Haque et al.	Airborne PM and respiratory health in Bangladesh: narrative review	<i>Health Science Reports</i>	Narrative review	PM _{2.5} , PM ₁₀ , UFP, gaseous co-pollutants	Bangladesh
Solazzo et al.	Season-by-pollutant interaction on the nasal microbiota of healthy adults	<i>Hygiene & Environmental Health Advances</i>	Repeated-measures biomarker	Indoor TSP; outdoor PM, BC, benzene, CO	Italy
Occupational & indoor					
Kaur et al.	Workplace graphene aerosol: personal EC exposure paired with in vitro lung toxicity	<i>NanoImpact</i>	Exposure assessment + in vitro	Airborne graphene / elemental carbon	Graphene production facility (Europe)
Nath et al.	Operating-room PM _{2.5} /PM ₁₀ and the effect of HEPA filtration and cautery	<i>J Anaesthesiol Clin Pharmacol</i>	Environmental surveillance campaign	PM _{2.5} , PM ₁₀ , CO ₂ , TVOC, HCHO	India
Thies et al.	London Underground commuting, cumulative PM _{2.5} and long-term mortality	<i>Int J Population Data Science</i>	Retrospective cohort	PM _{2.5} (metro, iron-rich; modelled cumulative)	London, UK
Sensing, forecasting & instrumentation					
Barkjohn et al.	Air Sensor Network Analysis Tool: an R-Shiny application for sensor QA/QC and reference comparison	<i>Atmosphere</i>	Software / tool description	PM _{2.5} , O ₃ (low-cost sensor vs regulatory reference)	United States
Burke et al.	Polar-grid nonparametric trajectory analysis for near-source pollutant attribution	<i>Atmospheric Environment</i>	Methods / near-road field data	PM _{2.5} , BC, NO _x	St. Louis, MO, USA
Chyzykhov et al.	Particle-bound water and its bias on gravimetric PM mass	<i>J Hazardous Materials</i>	Laboratory metrology	PM constituents, artificial PM, ambient PM1 filters	Laboratory
Ginsburg et al.	Transferability of US PurpleAir correction equations worldwide; regional recalibration	<i>Aerosol Science & Technology</i>	Multi-site co-location evaluation	PM _{2.5} (PurpleAir vs reference)	Global, 6 continents
Guo et al.	Mie-theory optical calibration plus aerosol liquid water separation for RH correction	<i>ACS ES&T Air</i>	Mechanistic correction framework	PM _{2.5} (nephelometer + OPC vs EPA reference)	United States (EPA CSN sites)
Kumpika et al.	Joint humidity and sensor-aging correction for PMS7003 light-scattering sensors	<i>Environ Pollutants & Bioavailability</i>	Field co-location + chamber	PM _{2.5} (PMS7003 vs BAM)	Chiang Mai, Thailand
Oybek et al.	Chamber quantification of T/RH sensitivity across three low-cost sensor chemistries	<i>Am J Applied Science & Technology</i>	Controlled chamber factorial	PM _{2.5} (OPC), NO ₂ (MOx), CO ₂ (electrochemical)	Laboratory
Qian et al.	AutoML multi-stage calibration for indoor low-cost PM _{2.5} sensors	<i>Atmospheric Measurement Techniques</i>	Chamber co-location, AutoML calibration	PM _{2.5} (indoor low-cost sensors)	Indoor chamber
Reis	Low-power sensor node with on-board quantised neural inference and adaptive LoRaWAN	<i>Sensors</i>	Simulation (system-level)	PM, gas, T/RH (embedded node)	Simulation
Yan	Review of machine-learning calibration for low-cost air quality sensor networks	<i>Sensor Review</i>	Narrative review	PM, O ₃ , NO ₂ (optical, electrochemical, MOx, NDIR)	Global
Yaqoob et al.	Multi-city bi-pollutant ML calibration of PurpleAir and Alphasense OPC-N3	<i>IEEE Access</i>	Calibration model comparison	PM _{2.5} , PM ₁₀ (PurpleAir, OPC-N3 vs TEOM/FIDAS)	Antwerp, Oslo, Zagreb
Zhivkov et al.	Distance-dependent calibration-transfer uncertainty in a hybrid reference/low-cost network	<i>Atmosphere</i>	Spatial cross-validation, operational network	PM _{2.5} , PM ₁₀ , NO ₂ , O ₃ (AirThings vs EEA reference)	Sofia, Bulgaria
Exposure assessment & modelling					
Ding et al.	Urban-rural balanced PM _{2.5} estimation corrects station-density bias	<i>Geo-spatial Information Science</i>	Machine-learning model	PM _{2.5} from AOD + reanalysis + ground	China, 2015-2021
Hu et al.	Hourly 1-km seamless PM _{2.5} from ground, satellite AOD and WRF-Chem fusion	<i>Geo-spatial Information Science</i>	Machine-learning model	PM _{2.5} , AOD, WRF-Chem	China
Kim et al.	Wildfire emissions suppress downwind anthropogenic sulfate via OH scavenging	<i>Environmental Pollution</i>	CTM simulation + source attribution	PM _{2.5} sulfate	Northeast Asia (Siberian fire, Jul 2014)
Lu et al.	Moving-vehicle wakes and noise-barrier interaction in traffic emission dispersion	<i>Atmospheric Environment</i>	Metadata only (no abstract)	Traffic emissions / near-road dispersion	Not stated in deposit
Nguyen et al.	Systematic review of satellite-plus-ML PM _{2.5} prediction and explainable AI	<i>Engineering Research Express</i>	Systematic review (PRISMA)	AOD, TOA reflectance, Angstrom exponent -> PM _{2.5}	Global

First author	Focus	Journal	Design	Metric	Region
Parajuli et al.	Soil NO/HONO parameterisation choice and its effect on modelled O3 and PM	<i>Atmospheric Environment</i>	Model intercomparison	PM _{2.5} , O3, NO2 (CMAQ)	Continental United States
Roky et al.	Distance-resolved Pb dispersion and crop uptake from open battery-plate burning	<i>Environmental Pollution</i>	Dispersion model + soil sampling	PM _{2.5} , PM ₁₀ , Pb deposition	Kushtia, Bangladesh
Yu et al.	Street-canyon increment to near-road NO2 quantified citywide with WRF-CMAQ-ADMS	<i>Environmental Pollution</i>	Chemical transport model	NO2 (traffic-related; canyon increment)	Hong Kong
Burden, policy & mitigation					
Gardner-Frolick et al.	Double disproportionality: concentrated PM _{2.5} emitters and marginalised siting in Canada	<i>iScience</i>	Spatial econometric model	PM _{2.5} and PM _{2.5} precursor emissions (NPRI)	Canada
Kim et al.	Spatial equity and cost-effectiveness of full diesel-to-BEV replacement	<i>J Environmental Management</i>	Chemical transport model	PM _{2.5} , O3 (population-weighted concentration)	Republic of Korea
Other clinical endpoints					
DeAngelis et al.	Residential context and epigenetic age acceleration: a null for air pollution	<i>Social Science & Medicine</i>	Cross-sectional (multistage)	Neighbourhood air pollution (1 of 5 contextual features)	United States
Oppebeck et al.	Exposome-wide search for drivers of epigenetic ageing, and the size of the missing exposome	<i>Environment International</i>	Exposome-wide association	Aerial black carbon among 186 exposures	Finland (FinnTwin12)

Provenance and method

Windows — and they are not uniform, which is stated rather than hidden. The entry-dated legs cover **2 August 2026** alone, contiguous with the 1 August issue, no gap. The harvest was executed over a 2–3 August query window and then **split on the [EDAT] value read from each record's own PubMed history** rather than on the query window: **18 of the 19 retained PubMed records carry an entry date of 2 August** and are carried here; one — Nimmala et al., transgenic switchgrass pollen dispersal, PMID 42543389 — carries **3 August** and is held back for that issue, logged to `state/rejected.jsonl` as HELD BACK rather than as a rejection so it cannot be lost. PubMed was queried along two independent axes: the health axis returned 21 records via the connector against 14 from the local harvester (the connector's larger yield is why both are run), the instrumentation axis returned 2, both already inside the health set. Union 23, 19 retained, 18 carried. Europe PMC CREATION_DATE: [2026-08-02 TO 2026-08-02] returned 0. The Crossref by-journal leg over eight sensing ISSNs returned 1 new deposit (Lu et al., *Atmos Environ*, created 2 August), carried metadata-only because Elsevier deposits no abstract. arXiv's relevance query returned nothing inside the recency screen — newest hit was 29 July, already past. OpenAlex returned **HTTP 429** on top of the standing paid-plan block on `from_created_date`; that leg remains dead and coverage is routed through Consensus as prescribed.

The out-of-window sensing sweep. A single-Sunday window on a literature PubMed structurally under-indexes would have produced a near-empty instrumentation section. Four Consensus sweeps — calibration and drift against reference monitors, satellite–surface fusion, network siting and edge inference, humidity and hygroscopic-growth correction — returned **12 records**, all new against `state/seen.json`, all in journals PubMed does not index (*Atmos. Meas. Tech.*, *Aerosol Sci. Technol.*, *ACS ES&T Air*, *IEEE Access*, *Geo-spatial Inf. Sci.*, *Sensor Review*, *Sensors*, *Atmosphere*). **These carry 2026 publication dates from January to June and are not inside the 2 August entry window.** Consensus exposes no entry-date filter, so they are screened by identity, not by date, and are dated to this issue because this is when the sweep surfaced them. **Read the 12-record sensing count as a backfill, not as a one-day flux** — any trend line drawn through it, including in the next weekly, will overstate the rate. DOIs were recovered by Crossref bibliographic matching, since Consensus returns none.

Screening. 31 records retained; **4 rejected with a logged reason** in `state/rejected.jsonl` — one urban-smellscape plant-VOC study with no particulate component, one drug-delivery nanocarrier paper in which PM_{2.5} appears only as a motivating sentence, one animal-free redox assay in keratinocytes with no inhalation route, and the *Sci Total Environ* retraction notice, which carries no research content and is noted in the editorial above instead. A fifth record is logged in the same file as HELD BACK rather than rejected — it is in scope and will be summarised in the 3 August issue on its own entry date. No cross-source duplicates arose; all 23 PubMed records and all 12 Consensus records were new against `seen.json` on PMID, DOI and title signature.

Trial watch. The ClinicalTrials.gov sweep on filtration, purifier, cookstove and mask interventions returned 25 active interventional records, of which **14 were new to state/trials.json** — among them the 770-participant multi-centre sham-controlled Clean Air RCT in COPD (NCT06376994), a 400-participant portable-air-cleaner heart-failure trial (NCT06070428), and two controlled woodsmoke-exposure chamber studies (NCT07536178, NCT06749093). They are surfaced in the weekly rollout, not here.

Subtopic, design, particle-metric and geography labels were assigned from title, abstract, keywords and MeSH terms; assignment is single-label by dominant contribution and is a judgement, not a controlled vocabulary. First-author attributions were verified against the PubMed `efetch` author lists rather than taken from summary metadata. Every DOI was verified and resolved before publication (0 fail, 6 advisory title-paraphrase warnings). Content derived from PubMed, Crossref and Consensus metadata; abstracts remain the copyright of their respective publishers.